THE JASMIN-CGN CORPUS

Design, recording, transcription and structure of the corpus

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1 Introduction

The aim of JASMIN-CGN project was to extend the Spoken Dutch Corpus (Corpus Gesproken Nederlands –CGN- a corpus of about 9 million words that constitutes a plausible sample of standard Dutch as spoken in the Netherlands and Flanders and contains various annotation layers) in three dimensions. First, by collecting a corpus of contemporary Dutch as spoken by children of different age groups, non-natives with different mother tongues and elderly people in the Netherlands and Flanders an extension along the age and mother tongue dimensions was achieved. In addition, speech material was collected in a communication setting that was not envisaged in the CGN: human-machine interaction. These three dimensions are reflected in the corpus as five user groups: native primary school pupils, native secondary school students, non-native children, non-native adults and senior citizens. For each group, both read and human-machine-interaction data had to be collected for a total of about 95 hours. One third of the data was to be collected in Flanders and two thirds in the Netherlands.

The aim of the project was to collect 95 hours of speech roughly divided as follows:
In The Netherlands:

- native children between 7 and 11 (12h 21m)
- native children between 12 and 16 (12h 21m)
- non-native children between 7 and 16 (12h 21m)
- non-native adults (12h 21 m)
- native adults above 65 (9h 26m)

In Flanders:

- native children between 7 and 11 (6h 10m)
- native children between 12 and 16 (6h 10m)
- non-native children between 7 and 16 (6h 10m
- non-native adults (6h 10m)
- native adults above 65 (5h 5m)

About 50% of the material would be read speech and 50% extemporaneous speech recorded in the human-machine-interaction modality. The amounts of material mentioned above together with the corresponding annotations also constituted the benchmarks for the JASMIN-CGN corpus. These amounts were considered as minimal targets which could of course be exceeded, but which had at least to be achieved.

2 Speaker selection

For the selection of speakers we have taken the following variables into account: region of origin (Flanders or the Netherlands), nativeness (native as opposed to non-native speakers), dialect region, age, gender and proficiency level in Dutch (in the case of non-native speakers).
In the **Flemish region** we finally recorded 207 speakers.
In the **Dutch region** we finally recorded 302 speakers.

2.1 Selection variable 1: region of origin

We distinguished two regions: Flanders (FL) and the Netherlands (NL) and we tried to collect one third of the speech material from speakers in Flanders and two thirds from speakers in the Netherlands.

2.2 Criterion 2: nativeness (native versus non-native speakers)

In each of the two regions, three groups of speakers consisted of native speakers of Dutch and two of non-native speakers. For native and non-native speakers different selection criteria were applied, as will be explained below.

2.3 Criterion 3: dialect region

Native speakers, on the other hand, were selected on the basis of the dialect region they belong to. A person is said to belong to a certain dialect region if (s)he has lived in that region between the ages of 3 and 18 and if (s)he has not moved out of that region more than three years before the time of the recording.
Within the native speaker categories we strived for a balanced distribution of speakers across the four regions (one core, one transitional and two peripheral regions) that we distinguished in the Netherlands and Flanders (see below), but without considering this as a hard demand. We adopted the same regions that were also used for CGN (Spoken Dutch) corpus.

In Flanders we distinguish the following four dialect regions:

1. FL1: West-Flemish (peripheral region): West-Flanders
2. FL2: East-Flemish (transitional region): East-Flanders
3. FL3: Brabant (core region): Antwerp and Flemish Brabant
4. FL4: Limburg (peripheral region): Limburg

In the Netherlands we distinguished the following four dialect regions:

1. West-Dutch (core region)
    a. N1a: South-Holland, excl. Goeree Overflakee
    b. N1b: North-Holland, excl. West Friesland
    c. N1c: West Utrecht, incl. the city of Utrecht
2. Transitional region
    a. N2a: Zeeland, incl. Goeree Overflakee and Zeeland Flanders
    b. N2b: Eastern Utrecht, excl. the city of Utrecht
    c. N2c: Gelders river area, incl. Arnhem and Nijmegen
    d. N2d: Veluwe as far as the IJssel
    e. N2e: West Friesland
    f. N2f: Polders
3. Northern (peripheral region):
    a. N3a: Achterkhoek
    b. N3b: Overijssel
    c. N3c: Drenthe
    d. N3d: Groningen
    e. N3e: Friesland
4. Southern peripheral region
    a. N4a: Noord-Brabant
    b. N4b: Limburg

For non-native speakers, dialect region did not constitute a selection variable, since the regional dialect or variety of Dutch is not expected to have a significant influence on their pronunciation. However, we did notice a posteriori that the more proficient non-native children do exhibit dialectal influence (especially in Flanders due to the recruitment). For this reason, information on dialect region was indicated for non-native speakers when appropriate.

2.4 Criterion 4: mother tongue (for non-native speakers only)

Since the JASMIN-CGN corpus was collected for the aim of facilitating the development of speech-based applications for children, non-natives and elderly people, special attention was paid to selecting and recruiting speakers belonging to the group of potential users of such applications. In the case of non-native speakers the applications we had in mind were especially language learning applications because there is considerable demand for CALL (Computer Assisted Language Learning) products that can help making Dutch as a second language (L2) education more efficient. In selecting non-native speakers, mother tongue constituted an important variable because certain mother tongue groups are more represented than others in the Netherlands and Flanders. For instance, for Flanders we opted for Francophone speakers since they form a significant fraction of the population in Flemish schools, especially (but not exclusively) in major cities. A language learning application could address the school's concerns about the impacts on the level of the Dutch class. For adults, CALL applications can be useful for social promotion and integration and for complying with the bilingualism requirements associated with many jobs. Often, Francophone speakers are immigrants from other countries and have other languages as their mother tongue. Such speakers were also allowed in the sample.

In the Netherlands, on the other hand, the choice of the mother tongue groups turned out to be less straightforward and even subject to change over time. The original idea was to select speakers with Turkish and Moroccan Arabic as their mother tongue, to be recruited in regional education centres where they follow courses in Dutch L2. This choice was based on the fact that Turks and Moroccans constitute two of the four most substantial minority groups (Dagevos, Gijsberts and Van Praag, 2003), the other two being people from Surinam and the Dutch Antilles who generally speak Dutch and do not have to learn it when they immigrate to the Netherlands. However, it turned out that it was very difficult and time-consuming to recruit exclusively Turkish and Moroccan speakers. In addition, the introduction of a new immigration law that envisages new obligations with respect to learning Dutch for people from outside the EU, led to considerable changes in the whole Dutch L2 education landscape, in particular to a significant decrease in the proportion of L2 learners from outside the EU. Moreover, in this modified context, it was no longer so straightforward to imagine which mother tongue groups would be the most obvious candidates for using CALL and speech-based applications. After various consultations with experts in the field, we decided not to limit the selection of non-natives to Turkish and Moroccan speakers and opted for a miscellaneous group that more realistically reflects the situation in Dutch L2 classes.

2.5 Criterion 5: proficiency in Dutch (for non-native speakers only)

Since an important aim in collecting non-native speech material is that of developing language learning applications for education in Dutch L2, we consulted various experts in the field to find out for which proficiency level such applications are most needed. It turned out that for the lowest levels of the Common European Framework

(CEF), namely A1, A2 or B1 there is relatively little material and that ASR-based applications would be very welcome. For this reason, we chose to record speech form adult Dutch L2 learners at these lower proficiency levels. For further details on CEF, see:
http://taalunieversum.org/onderwijs/publicaties/gemeenschappelijk_europees_referentiekader/.
For children, the current class (grade) they are in was maintained as a selection criterion.

2.6 Criterion 6: age of the speaker

Age was used as a variable in selecting both native and non-native speakers. For the native speakers we distinguished three age groups:

- children between 7 and 11
- children between 12 and 16
- native adults of 65 and above

For the non-native speakers two groups were distinguished:

- children between 7 and 16
- adults between 18 and 60

2.7 Criterion 7: gender of the speaker

In the five groups of speakers we strived to obtain a balanced distribution between male and female speakers.

3 Selection of speech material

In order to obtain a relatively representative and balanced corpus we decided to record about 12 minutes of speech from each speaker. About 50% of the material would consist of read speech material and 50% of extemporaneous speech produced in human-machine dialogues.

3.1 Read speech material

About half of the material to be recorded from each speaker in this corpus consists of read speech. For this purpose we used sets of phonetically rich sentences and stories or general texts to be read aloud. Particular demands on the texts to be selected were imposed by the fact that we had to record read speech of children and non-natives.
Children in the age group 7-12 cannot be expected to be able to read a text of arbitrary level of difficulty. In many elementary schools in the Netherlands and Flanders children learning to read are first exposed to a considerable amount of explicit phonics instruction which is aimed at teaching them the basic structure of written language by showing the relationship between graphemes and phonemes (Wentink, 1997). A much used method for this purpose is the reading program Veilig Leren Lezen (Mommers et al., 1990). In this program children learn to read texts of increasing difficulty levels, with respect to text structure, vocabulary and length of words and sentences. The texts are ordered according to reading level and they vary from Level 1 up to Level 9.
In line with this practice in schools, we selected texts of the nine different reading levels from books that belong to the reading programme Veilig Leren Lezen.
For the non-native speakers we selected appropriate texts from books that are intended for learners of Dutch as a second language (Dutch L2). Owing to IPR restrictions we could not include the complete texts in the documentation.

3.2 Human-machine dialogues

A WoZ-based platform was developed for recording speech in the human-machine interaction mode. The human-machine dialogues are designed such that the wizard can intervene when the dialogue goes out of hand. In addition, the wizard can simulate recognition errors to elicit some of the typical phenomena of human-machine interaction that are known to be problematic in the development of spoken dialogue systems. Before designing the dialogues we drew up a list of phenomena that should be elicited such hyperarticulation, syllable lengthening, shouting, stress shift, restarts, filled pauses, silent pauses, self talk, talking to the machine, repetitions, prompt/question repeating and paraphrasing

We then considered which speaker's moods could cause the various phenomena and identified three relevant states of mind: (1) confusion, (2) hesitation and (3) frustration. If the speaker is confused or puzzled, (s)he is likely to start complaining about the fact that (s)he does not understand what to do. Consequently, (s)he will probably start talking to him/herself or to the machine. Filled pauses, silent pauses, repetitions, lengthening and restarts are likely to be produced when the speaker has doubts about what to do next and looks for ways of taking time. So hesitation is probably the state of mind that causes these phenomena. Finally, phenomena such as hyperarticulation, syllable lengthening, syllable insertion, shouting, stress shift and self talk probably result when speakers get frustrated. As is clear from this characterization, certain phenomena can be caused by more than one state of mind, like self talk that can result either from confusion or from frustration.

The challenge in designing the dialogues was then how to induce these states of mind in the speakers, to cause them to produce the phenomena required.

We have achieved this in different ways such as asking unclear questions, increasing the cognitive load of the speaker by asking more difficult questions, or simulating machine recognition errors of different.

Different dialogues were developed for the different speaker groups. To be more precise, the structure was similar for all the dialogues, but the topics and the questions were different.

4. Speaker recruitment

Different recruitment strategies for native and non-native speakers and for children and adult speakers were applied. There has been some cooperation with the AUTONOMATA project, which also recorded young and non-native speakers.

4.1 Recruitment of native children

<u>Research institutes and schools</u> 教学研究机构

The original aim was to recruit children through pedagogical research institutes that have regular access to schools for various experiments. This seemed to be the most viable solution in the Netherlands, because schools are reluctant to participate in individual projects owing to a general lack of time. Unfortunately, this form of mediation turned out not to work because pedagogical institutes give priority to their own projects. So, eventually, we had to contact the schools ourselves. For these reasons, recruiting children turned out to be much more time-consuming than we had envisaged.

In Flanders, most recordings were done at schools by seeking collaboration with the management of the schools. A small fraction of the data were recorded at summer recreational activities for primary school children ("speelpleinwerking")

Friends and family
In certain cases we had to resort to friends and family members to find enough speakers, who, in turn, were asked whether they knew other potential candidates who would be willing to participate.

4.2 Recruitment of native elderly adults

The elderly people were recruited through retirement homes and elderly care homes. In Flanders older adults were also recruited through a Third Age University.

4.3 Recruitment of non-native children

The Netherlands
Several schools (ISKs (Internationale Schakelklassen) offer specific Dutch courses for foreign children who immigrate to the Netherlands. Through a number of these schools we could contact students with the right level of proficiency in Dutch (above CEF level A1).

Flanders
In Flanders the non-native children were primarily recruited in regular schools. In major cities and close to the language border a significant proportion of pupils speak only French at home but attend Flemish schools. The level of proficiency is very dependent on the individual and the age.
A second source of speakers was a school with special programs for recent immigrants.

4.4 Recruitment of non-native adults

Language Schools
Several schools (in the Netherlands Regionale Opleidingscentra, ROCs – in Flanders Centra voor VolwassenenOnderwijs, CVO's) offer Dutch courses for foreigners. Through these schools we managed to contact non-native speakers with the right level of linguistic skills.

Organizations
Specific organizations for foreigners were also contacted to find enough speakers when recruitment through the schools failed.

4.5 Speaker forms

Every speaker had to fill in a speaker form in which he/she declares that the recordings may be used for scientific and limited commercial purposes. One of the parents was asked to sign the form for speakers that had not reached the age of 18 at the time of recording.

The speaker form also contained the following personal details which were then recorded in the speaker database:

1. Name, Address, Phone, E-mail
2. Nationality
3. Place of birth
4. Dialect area: the city where they have lived between the ages of 3 and 18 (or to date for children)

5. Number of years in the region (for non-native speakers)
6. Language spoken at home
7. Education level (for adult natives) or CEF level (for non-native adults)
8. Birth date
9. Gender

Obviously, in the final distribution of the speaker data, names have been replaced by a speaker code and no address information is released. Unfortunately, there are some missing forms and, as a consequence, some missing data. In particular, we miss the speaker forms for the group 2 speakers in the region with ZIP code N-1403. The parents of these children did give permission for the recordings and the forms were collected by one teacher, who subsequently lost them in a mysterious way.

5 Making the recordings

This section contains information about how and where the different recordings were made and how the data were stored in the computer.

5.1 Speech elicitation

To record read speech, the speakers were asked to read texts that appeared on the screen. They were instructed to read at a normal tempo and were given the opportunity to practice before starting the recordings proper. To elicit speech in the human-machine interaction modality, on the other hand, the speakers were asked to have a dialogue with the computer. They were asked questions that they could also read on the screen and they had received instructions that they could answer these questions freely and that they could speak as long as they wanted.

5.2 Recording locations

The recordings were made on location in schools and retirement homes. We always tried to obtain a quiet room for the recordings. Nevertheless, background noise and reverberation could not always be prevented.

5.3 Audio equipment

The recording platform consists of four components: the microphone, the amplifier, the soundcard and the recording software. We used a Sennheiser 835 cardoid microphone to limit the impact of ambient sound. The amplifier is integrated in the soundcard (M-audio) and contains all options for adjusting gain en phantom power.
Resolution is 16bit, which is considered sufficient according to the CGN specifications.
The microphone and the amplifier are separated from the PC, so as to avoid interference between the power supply and the recordings.
Elicitation techniques and recording platform were specifically developed for the JASMIN-CGN project because one of the aims was to record speech in the human-machine-interaction modality. The recordings are stereo, as both the machine output and the speaker output are recorded.

5.4 Sampling and encoding

The samples were stored in 16 bit linear PCM form, uncompressed in a Microsoft Wave Format. The sample frequency was 16 kHz for all recordings. The dialogue recording contain two channels: the output from the TTS system and the microphone recording. Notice that the microphone signal also contains the TTS signal through the acoustic path from the loudspeakers to the microphone. Read speech recordings contain one channel.

5.5 Recording sessions

Before the actual recording was started, some test recordings were performed to check whether all scripts were installed correctly, whether the settings of the amplifier were OK and whether the placement of the microphone was OK.

5.6 Duration measurement

The amount of material to be recorded is expressed in temporal units (as opposed to number of words, for instance). The duration of a recording is the "active turn" time of the speaker. This starts when the prompt of the dialogue system starts and ends when the speaker stops talking. The reaction time of the wizard and the startup time of the session are excluded. The prompt time is included because speakers also speak through the prompt as a discourse act. These phenomena are also annotated.
For read speech, the recording time is counted from the first word of the speaker till the last one.

5.7 Compensation for participants

All speakers received a small compensation for participating in the recordings in the form of a cinema ticket or a coupon for a bookstore or a toy store.

6 Orthographic transcription of the recordings

6.1 Transcription procedure

All speech recordings were orthographically transcribed according to the same conventions adopted in CGN by using the PRAAT program. Transcribers were selected form a pool of trained transcribers that had already worked for CGN or for similar projects. Since this corpus also contains speech by non-native speakers, special conventions were required, for instance, for transcribing words realised with non-native pronunciation. The orthographic transcriptions are made according to the protocol: corpus_orth-transcriptie.pdf and are in ISO8859 format.

6.2 Quality of the transcriptions

After the orthographic transcriptions have been made by one transcriber, they are checked by a second transcriber who listens to the sound files, checks whether the orthographic transcription is correct and, if necessary, improves the transcription. A spelling check is also carried out according to the latest version of the Dutch spelling (*Woordenlijst Nederlandse Taal*, 2005). A final check on the quality of the orthographic transcription is carried out by running the program 'orttool'. This program, which was developed for CGN, checks whether markers and blanks have been placed correctly and, if necessary, improves the transcription.
The speech material recorded in the Netherlands was also transcribed in the Netherlands, whereas the speech material recorded in Flanders was transcribed in Flanders. To prevent inconsistencies in the transcription, cross checks were carried out.

7 Transcription of Human-Machine-Interaction phenomena

7.1 Transcription procedure

A new protocol was drawn up for transcribing the HMI phenomena that were elicited in the dialogues. The aim of this type of annotation was to indicate these phenomena so that they can be made accessible for all sorts of research and modeling. As in any type of annotation, achieving an acceptable degree of reliability is very important. For this reason in the protocol we identified a list of phenomena that appear to be easily observable and that are amenable to subjective interpretation as little as possible. In addition, examples were provided of the manifestation of these phenomena, so as to minimize subjectivity in the annotation.

7.2 Quality of the transcriptions

As for the orthographic transcriptions, the HMI transcriptions were also made by one transcriber and checked by a second transcriber who listened to the sound files, checked whether the transcription was correct and, if necessary, improved it. The speech material recorded in the Netherlands was also transcribed in the Netherlands, whereas the speech material recorded in the Flanders was transcribed in Flanders. To prevent inconsistencies in the transcription, cross checks were carried out.

8 Phonemic transcription of the recordings

It is common knowledge, and the experience gained in CGN confirmed this, that manually generated phonetic transcriptions are very costly. In addition, recent research findings indicate that manually generated phonetic transcriptions are not always of general use and that they can be generated automatically without considerable loss of information (Van Bael et al., 2003). In a project like JASMIN-CGN then an important choice to make is whether the money should be allocated to producing more detailed and more accurate annotations or simply to collecting more speech material. Based on the considerations mentioned above and the limited budget that was available for collecting speech of different groups of speakers, we chose the second option and decided to adopt an automatically generated broad phonetic transcription (using Viterbi alignment). For this purpose, the CGN conventions were adopted, as well as the same symbol set. The annotation files are in ISO8859 format.

8.1 Transcription procedure

Given the nature of the data (non-native, different age groups and partly spontaneous), the procedure requires some care. Since the performance of an automatic speech aligner largely depends on the suitability of its acoustic models to model the data set, it was necessary to divide the data into several categories and treat each of those separately. In all cases, we started out from an *initial* acoustic model and adapted that to each category by mixing in the data on which we need to align. Moreover, for children, we applied a vocal tract length compensation by grouping them in 3 age groups (5-9 years old, 10-12 years old and 13-16 years old) and determining the optimal warping factor per group.

We used four initial acoustic models: Dutch native children (trained on roughly 14 hours of JASMIN data), Flemish native children (trained on a separate database), Dutch native adults (trained on CGN) and Flemish native adults (trained on several

separate databases). Each data set can be phonetically annotated in the same way using one of these models. For instance, to align the Flemish senior speech, a single training pass was performed on the model for Flemish native adult speech using the Flemish senior data.
The quality of the automatic annotation obtained by the speech aligner depends on the quality of the lexicon used. These lexicons should contain pronunciation variants as options the Viterbi aligner can choose from. For instance, the "n" can be omitted in certain contexts. The base lexica were Fonilex for Flemish and CGN for Dutch. Two pronunciation phenomena were additionally annotated manually in the JASMIN-database: pause in a word (*s) and foreign pronunciation of a word (*f). The lexicon for these words was created manually in several iterations of inspection and lexicon adaptation. In general, this leads to an increase in the options the Viterbi aligner can choose from. Further modelling of pronunciation variation is in hard-coded rules as in the CGN. Additional information on the phonemic annotation can be found in : C. Cucchiarini, J. Driesen, H. Van hamme and E. Sanders (2008) Recording Speech of Children, Non-Natives and Elderly People for HLT Applications: the JASMIN-CGN Corpus, LREC 2008, Marrakech, Morocco.

8.2 Quality of the transcriptions

The quality of the automatically generated phonemic transcriptions was verified for 3 randomly selected files per Region (FL/NL) and category (non-native child, non-native adult, native child and senior) (a total of 24 recordings) by inspection of the proposed transcription. Lexicon and cross-word assimilation rules were adapted to minimize the number of errors. Most of the required corrections involved hard/soft pronunciation of the "g" and optional "n" in noun plurals and infinitive forms.

9 Part-of-speech tagging

For all (orthographic) transcriptions, a part of speech (POS) tagging was made. This was done fully automatically by using the POS tagger that was developed for CGN at ILK/Tilburg University. Accuracy of the automatic tagger in CGN was about 95% (Van den Bosch et al. 2006).
The tagset consists of 316 tags and is extensively described (in Dutch) in:
Van Eynde (2004).
POS tagger and documentation are described in:
Van den Bosch, A., Busser, G.J., Canisius, S., and Daelemans, W. (2007).
Manual correction of the automatic POS tagging was not envisaged in this project.
One part of a speech file is missing (fv160056.pos). This file cannot be reproduced.

10 Structure of the corpus

In this section we discuss the organization of the corpus (directory structure, filename conventions, etc.) and the format and content of the data and metadata files.

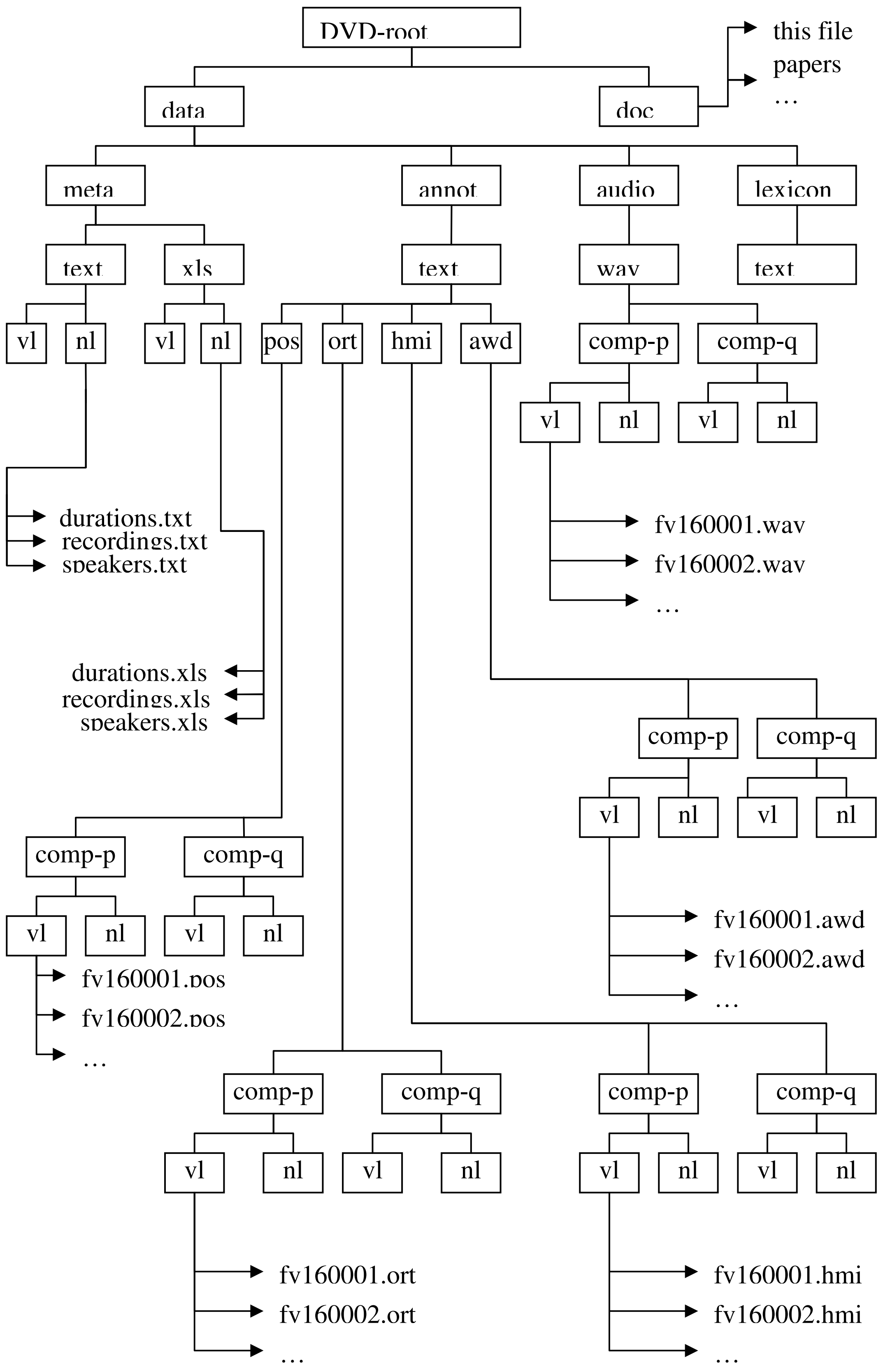
10.1 Directory structure and file nomenclature

The data are organized such that it follows the directory structure of the CGN corpus as closely as possible. The data are organized as two additional components in the CGN corpus: comp-p (dialogues) and comp-q (read). Comp-p is obviously a new mode along the dimension "components" of the CGN corpus, as human-machine interaction was not part of the corpus before. For the read texts, it was preferred to add a new component since the nature of the texts differs from CGN comp-o as well to separate it clearly from the original CGN data. However, there are other important differences with CGN, such as age and non-native speech. Hence, other data organizations can be proposed and defended. To facilitate other data organizations and to allow users to easily select the subset they are interested in, the recordings.xls (and recordings.txt) and speakers.xls (and speakers.txt) files both include information such as the target group (senior, non-native child, etc.) (see below). Because of the new dimensions introduced in the JASMIN corpus, the formats for the recordings.xls (and .txt) and speaker.xls (and .txt) of the CGN corpus could not be maintained.

Like in the CGN corpus each new component further contains a subdirectory “nl” and “vl” for “the Netherlands” and “Flanders”. The marginal distributions of the total recording time as defined in section 5.6 is given in the files duration.xls (and duration.txt) for each group (1 through 5 meaning: native children 7-11, native children 12-16, non-native children, non-native adults and native adults above 65, respectively).

The file name root (e.g. fv160004) consists of f, followed by either n (Netherlands) or v (Vlaanderen, Flanders) and a six-digit code which is automatically created by the recording platform. Used file name extensions are .wav, .awd, .ort, .hmi and .pos.

Directory tree:



The files “recordings” and “speakers” (.txt => text format; .xls => MS Excel) describe the recording info (o.a. which speaker) and the speaker characteristics respectively. This way, a user can link a file name (eg fv160001) with a speaker.

10.2 Speech and transcription file formats

De *speech files* in the dialogue component (comp-p) are in stereo (one channel for the microphone signal and one channel for the TTS prompt) while the speech files in the read speech part (comp-q) are mono. All files are delivered in Microsoft Wave Format.

De *transcription files* are delivered as text files in UNIX format. For convenience and compatibility with CGN, the following files are provided:

.ort: orthography containing tiers: orthography of TTS channel, orthography of microphone channel, background and comment

.hmi: orthography of TTS channel, orthography of microphone channel, HMI events, dialog state, avatar events, screen events, background, comment

.awd: orthography of microphone channel, word boundaries with phonemic transcription and phoneme boundaries with phonemic annotations (in line with CGN conventions, see:
http://lands.let.ru.nl/cgn/doc_English/topics/version_1.0/formats/text/awd.htm).

The pos files contain the parts-of speech tags.

The phonetic alphabet and the transcription conventions are presented in the protocols included in the documentation.

10.3 Lexicon

The lexicon of all words that occur in JASMIN-CGN is provided in data/lexicon/text. The file named "lexicon_sgml.txt" follows the CGN format as described in http://lands.let.ru.nl/cgn/doc_English/topics/version_1.0/lexicon/lexicon.htm, which indicates the fields and the phonemic alphabet used. The file named lexicon_iso.txt is an ISO-8859 encoding of the same lexicon and is in line with the format of the annotations. In JASMIN-CGN, only the orthography (2nd field), the “lexicon status” (7th field) and the phonemic transcription (8th field for NL, 9th field for VL) are provided. The 7th field contains the extended lexicon codes as described in section B.4 of the JASMIN protocol for orthographic transcription. The phonemic transcriptions in the 8th and 9th fields are all the transcriptions that were found during the automatic alignment described in section 8.1. The alternatives are separated by slash “/”.

10.4 Speaker and recording information files

The speaker information available is provided in the form of a text file called **speakers.txt** and in the form of an excel-file **speakers.xls**. Each line in the text file is composed of semicolon separated items. The first line reveals the meaning of each item, whereas the subsequent lines provide the data of the subsequent speakers:

1. region (N/V) + speaker id (a 6 digit number): (e.g. V000128)
2. zipcode of current place of residence
3. gender (M or F, this differs from CGN that used sex1 and sex2)
4. age at the time of recording (in years)
5. zipcode or country code of place of birth (see countries.txt)
6. group
   1: native children aged 7-11
   2: native children aged 12-16
   3: non-native children

4: non-native adults
5: native adults above 65

7. code of language spoken at home (see languages.txt)
8. code of 2nd language spoken at home (see languages.txt)
9. CEF proficiency level of Dutch (for adult non-natives) (A1, A2 or B1)
10. comment code:
    1. "0" means "no comment"
    2. "1" is used for a speaker whose Northern Dutch accent dominates of the French
    3. "N" is used for children who speak French at home, but who are very proficient in Dutch
11. educational level(for speakers of age > 18) (1 … 4)
12. place of education (0-18years) (see countries.txt)
13. dialect region (for native speakers) (NL1-4, FL1-4)
14. length stay in NL or VL
15. time on Dutch L2

Owing to the large amount of variation in languages and countries in the speaker data, the CGN coding of languages and countries could not be applied in the JASMIN-CGN corpus because it contained an insufficient number of codes. As an alternative, the ISO codes for countries and languages were applied (see languages.txt and countries.txt and further http://en.wikipedia.org/wiki/ISO_3166-1_alpha-3 and ISO 639-2 Language Code List - Codes for the representation of names of languages (Library of Congress).mht). For coding the language variety spoken at home we created two new codes in the ISO style, dia for dialect and tus for "tussentaal" a regional variety in between standard language and dialect. Dialect regions have been included for non-native speakers when appropriate because they audibly affect speech production. Some non-natives are born in Flanders but raised in French and go to local schools where they are confronted with local dialects, which affects their speech.
CEF level was coded for the adult non-natives, but not for the non-native children because for this latter group reliable information in this respect could not be obtained. However, in selecting the children in schools teachers were instructed to select children that would at least be at level A1.
For educational level, the following codes apply both in The Netherlands and Flanders

1. Primary school
2. Secondary school
3. Higher education (short type up to 3 years)
4. Higher education longer type (more than 3 years)

Example text file
RegionSpeaker;ResPlace;Gender;Age;BirthPlace;Group;HomeLanguage1;HomeLanguage2;CEF;Comment;EduLevel;EduPlace;DialectRegion;Length stay in NL or VL; Time on Dutch L2
…
V000015;B-178;F;22;B-180;4;FR;FR;A1;0;3;B-178;;
…
V000089;B-351;M;7;B-350;1;VL;VL;;0;;B-351;FL4;

Of every recording, the following information is made available in recordings.xls and recordings.txt:

1. file name root (e.g. fv160004)
2. speaker id (e.g. V000128): refers to a line in speakers.txt
3. component (comp-p or comp-q)
4. group
   1: native children aged 7-11
   2: native children aged 12-16
   3: non-native children
   4: non-native adults
   5: seniors
5. Age
6. Gender (M or F, this differs from CGN that used sex1 and sex2)
7. CEF level
8. Dialect region
9. Duration (as defined above) in seconds
10. Duration (as defined above) in days (provided for processing in MS Excel)

Strictly speaking, columns 4 through 10 can be derived from the speaker information files. These columns are added such that users can easily select subgroups from the JASMIN-CGN corpus.
The recordings.txt specifies the relation between recording and speaker in every detail. In short: most speakers participated in one comp-p and one comp-q recording. Some participated in only one of both components. None contributed twice to the same recording. Which text is read/dialogue is executed in recordings.txt/xls.

10.5 Documentation

The documentation directory contains the following files:

- corpus_documentation.pdf: Corpus Documentation (in English) (this file)
- corpus_orth-transcription.pdf: Orthographic transcription protocol (in English)
- corpus_orth-transcriptie.pdf: Orthographic transcription protocol (in Dutch)
- corpus_HMI-transcription.pdf: HMI transcription protocol (in English)
- corpus_HMI-transcriptie.pdf: HMI transcription protocol (in Dutch)
- countries.txt
- languages.txt

10.6 Distribution media

The corpus is distributed on one CD and six DVDs and is about 23 GByte. The CD contains the documentation and the metadata, while the DVDs contain the speech data and the corresponding annotations, distributed as follows:
DVD01:
comp-p/nl, fn000029-fn000449 (75 spk)
comp-q/nl, fn000030-fn000450 (75 spk)
DVD02:
comp-p/nl, fn000452-fn000697 (75 spk)
comp-q/nl, fn000453-fn000698 (74 spk)
DVD03:
comp-p/nl, fn000699-fn000878 (75 spk)
comp-q/nl, fn000700-fn000879 (76 spk)

DVD04:
comp-p/nl, fn000880-fn100326 (75 spk)
comp-q/nl, fn000881-fn100327 (75 spk)
DVD05:
comp-p/vl, fv160000-fv160099 , fv170092-fv170095 (88 spk)
comp-q/vl, fv160014-fv160098 , fv170001-fv170099 (97 spk)
DVD06:
comp-p/vl, fv160105-fv160218 (105 spk)
comp-q/vl, fv160137-fv160213 , fv170100-fv170209 (105 spk)

10.7 References

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